MEDIUM ALGEBRA ANSWER KEY

Linear functions

02

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ANSWER KEY · SAT QUESTION BANK

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Q1

D

D. $fx = 4x + 8$

Choice D is correct. Since f is a linear function, it can be defined by an equation of the form $fx = ax + b$, where a and b are constants. It's given that $f0 = 8$. Substituting 0 for x and 8 for fx in the equation $fx = ax + b$ yields $8 = a0 + b$, or $8 = b$. Substituting 8 for b in the equation $fx = ax + b$ yields $fx = ax + 8$. It's given that $f1 = 12$. Substituting 1 for x and 12 for fx in the equation $fx = ax + 8$ yields $12 = a1 + 8$, or $12 = a + 8$. Subtracting 8 from both sides of this equation yields $a = 4$. Substituting 4 for a in the equation $fx = ax + 8$ yields $fx = 4x + 8$. Therefore, an equation that defines f is $fx = 4x + 8$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q2

B

B. 2

Choice B is correct. The value of $f(9) - f(1)$ can be calculated by finding the values of $f(9)$ and $f(1)$. The value of $f(9)$ can be found by substituting 9 for x in the given function: $f(9) = \frac{(9+7)}{4}$. This equation can be rewritten as $f(9) = \frac{16}{4}$, or 4. Then, the value of $f(1)$ can be found by substituting 1 for x in the given function: $f(1) = \frac{(1+7)}{4}$. This equation can be rewritten as $f(1) = \frac{8}{4}$, or 2. Therefore, $f(9) - f(1) = 4 - 2$, which is equivalent to 2.

Choices A, C, and D are incorrect and may result from incorrectly substituting values of x in the given function or making computational errors.

Q3

B

B. $fx = -2x + 7$

Choice B is correct. The graph of a line in the xy-plane can be represented by the equation $y = mx + b$, where m is the slope of the line and $0, b$ is the y-intercept. The slope of a line that passes through the points x_1, y_1 and x_2, y_2 can be calculated using the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. The line shown passes through the points $-1, -2$ and $0, -4$. Substituting $-1, -2$ and $0, 4$ for x_1, y_1 and x_2, y_2, respectively, in the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{-4 - -2}{0 - -1}$, which is equivalent to $m = \frac{-2}{1}$, or $m = -2$. Since the line shown passes through the point $0, -4$, it follows that $b = -4$. Substituting -2 for m and -4 for b in the equation $y = mx + b$ yields $y = -2x - 4$. It's given that the graph shown is the graph of $y = fx - 11$. Substituting $-2x - 4$ for y in the equation $y = fx - 11$ yields $-2x - 4 = fx - 11$. Adding 11 to both sides of this equation yields $-2x + 7 = fx$. Therefore, the equation $fx = -2x + 7$ defines the linear function f.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q4

B

B. 3.04

Choice B is correct. It's given that the function $f(x) = 55.20 - 0.16x$ gives the estimated surface water temperature, in degrees Celsius, of a body of water on the xth day of the year. Substituting 326 for x in the given function yields $f(326) = 55.20 - 0.16(326)$, which is equivalent to $f(326) = 55.20 - 52.16$, or $f(326) = 3.04$. Therefore, the estimated surface water temperature, in degrees Celsius, of this body of water on the 326th day of the year is 3.04.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the rate of change, in degrees Celsius per day, of the estimated surface water temperature.

Choice D is incorrect. This is the change, in degrees Celsius, in the estimated surface water temperature over 326 days.

Q5 257

The correct answer is 257. It’s given that a linear model estimates the population of a city from 1991 to 2015. Since the population can be estimated using a linear model, it follows that there is a constant rate of change for the model. It’s also given that the model estimates the population was 57 thousand in 1991, 224 thousand in 2011, and x thousand in 2015. The change in the population between 2011 and 1991 is 224 - 57, or 167, thousand. The change in the number of years between 2011 and 1991 is 2011 - 1991, or 20, years. Dividing 167 by 20 gives 167 / 20, or 8.35, thousand per year. Thus, the change in population per year from 1991 to 2015 estimated by the model is 8.35 thousand. The change in the number of years between 2015 and 2011 is 2015 - 2011, or 4, years. Multiplying the change in population per year by the change in number of years yields the increase in population from 2011 to 2015 estimated by the model: 8.354, or 33.4, thousand. Adding the change in population from 2011 to 2015 estimated by the model to the estimated population in 2011 yields the estimated population in 2015. Thus, the estimated population in 2015 is 33.4 + 224, or 257.4, thousand. Therefore to the nearest whole number, the value of x is 257.

Q6 2

The correct answer is 2. It’s given that function f is defined by $fx = 2x + 3$. Therefore, the equation representing the graph of $y = fx$ in the xy-plane is $y = 2x + 3$, and the graph is a line. For a linear equation in the form $y = mx + b$, m represents the slope of the line. Since the value of m in the equation $y = 2x + 3$ is 2, the slope of the line defined by function f is 2. It's given that line j is parallel to the line defined by function f. The slopes of parallel lines are equal. Therefore, the slope of line j is also 2.

Q7

C

C. Caleb had approximately 30 fluid ounces of juice when he began to make the popsicles.

Choice C is correct. An equation that defines a linear function f can be written in the form $fx = mx + b$, where m represents the slope and b represents the y-intercept, $0, b$, of the line of $y = fx$ in the xy-plane. The function $fx = -5x + 30$ is linear. Therefore, the graph of the given function $y = fx$ in the xy-plane has a y-intercept of $0, 30$. It's given that fx gives the approximate volume, in fluid ounces, of juice Caleb had remaining after making x popsicles. It follows that the y-intercept of $0, 30$ means that Caleb had approximately 30 fluid ounces of juice remaining after making 0 popsicles. In other words, Caleb had approximately 30 fluid ounces of juice when he began to make the popsicles.

Choice A is incorrect. This is an interpretation of the slope, rather than the y-intercept, of the graph of $y = fx$ in the xy-plane.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Q8

A

A. $P(t) = 862 - 1.6t$

Choice A is correct. It is given that the relationship between population and year is linear; therefore, the function that models the population t years after 2000 is of the form $P(t) = mt + b$, where m is the slope and b is the population when $t = 0$. In the year 2000, $t = 0$. Therefore, $b = 862$. The slope is given by $m = \frac{P(10) - P(0)}{10 - 0} = \frac{846 - 862}{10 - 0} = \frac{-16}{10} = -1.6$. Therefore, $P(t) = -1.6t + 862$, which is equivalent to the equation in choice A.

Choice B is incorrect and may be the result of incorrectly calculating the slope as just the change in the value of P. Choice C is incorrect and may be the result of the same error as in choice B, in addition to incorrectly using t to represent the year, instead of the number of years after 2000. Choice D is incorrect and may be the result of incorrectly using t to represent the year instead of the number of years after 2000.

Q9

A

A. 41

Choice A is correct. It's given that the equation $y = 0.67x + 2.6$, where $20 \leq x \leq 110$, gives the predicted number of larvae, y, in a colony of ants if the colony has x worker ants. If one of these colonies has 58 worker ants, the predicted number of larvae in that colony can be found by substituting 58 for x in the given equation. Substituting 58 for x in the given equation yields $y = 0.6758 + 2.6$, or $y = 41.46$. Of the given choices, 41 is closest to the predicted number of larvae in the colony.

Choice B is incorrect. This is closest to the predicted number of larvae in a colony with 87 worker ants.

Choice C is incorrect. This is closest to the number of worker ants for which the predicted number of larvae in a colony is 58.

Choice D is incorrect. This is closest to the predicted number of larvae in a colony with 280 worker ants.

Q10 D

D. $h\left(x\right) = -\frac{7}{2}x + 113$

Choice D is correct. An equation defining h can be written in the form $y = mx + b$, where $y = hx$, m represents the slope of the graph of $y = hx$ in the xy-plane, and b represents the y-coordinate of the y-intercept of the graph. It's given that $h28 = 15$ and $h26 = 22$. It follows that the points $28, 15$ and $26, 22$ are on the graph of $y = hx$ in the xy-plane. The slope can be found by using any two points, x_1, y_1 and x_2, y_2, and the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting $28, 15$ and $26, 22$ for x_1, y_1 and x_2, y_2, respectively, in the slope formula yields $m = \frac{22 - 15}{26 - 28}$, or $m = -\frac{7}{2}$. Substituting $-\frac{7}{2}$ for m and $28, 15$ for x, y in the equation $y = mx + b$ yields $15 = -\frac{7}{2}28 + b$, or $15 = -98 + b$. Adding 98 to both sides of this equation yields $113 = b$. Substituting $-\frac{7}{2}$ for m and 113 for b in the equation $y = mx + b$ yields $y = -\frac{7}{2}x + 113$. Since $y = hx$, it follows that the equation that defines h is $hx = -\frac{7}{2}x + 113$.

Choice A is incorrect. For this function, $h26 = \frac{109}{7}$, not $h26 = 22$.

Choice B is incorrect. For this function, $h28 = 105$, not $h28 = 15$, and $h26 = \frac{739}{7}$, not $h26 = 22$.

Choice C is incorrect. For this function, $h28 = -75$, not $h28 = 15$, and $h26 = -68$, not $h26 = 22$.

Notes

Accept equivalent forms (e.g. `0.25` ≡ `1/4`). For grid-in items, decimal and fraction answers are both valid.

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